

PhysMon: Next-Phase Scientific and Experimental Plan

Part II - From Monitoring Evidence to a Defensible Causal Account

Continuation of

“PhysMon: Hidden-State Detection of Physics Shortcut Sensitivity under Solver-Verified Counterfactuals”

Document purpose. This document converts the completed and ongoing PhysMon evidence into a structured next-phase research programme. It specifies the validation work, confirmatory analyses, causal experiments, mechanistic deepening, external-validity studies, mitigation experiments, decision gates, and reproducibility standards required to produce the strongest scientifically defensible paper possible.

Research planning version 1.1
Revised June 22, 2026

Status: active continuation document; designed to be updated through controlled versioning

Relationship to Part I. Part I defined the construct, benchmark, model roles, hidden-state monitoring methodology, causal scope, artefact controls, and initial decision gates. Part II assumes that the initial benchmark, behavioural sweeps, probing studies, and causal localization experiments have been executed. It does not replace Part I. It governs the next research phase: evidence stabilization, independent validation, stronger confirmation, deeper mechanism identification, generalization, and selective correction.

Document Control

Field	Specification
Document name	PhysMon: Next-Phase Scientific and Experimental Plan, Part II
Governing predecessor	PhysMon Research Proposal and Supplementary Documentation, June 13, 2026
Primary purpose	Define the complete next-phase programme required to move from promising monitoring and localization evidence to a rigorous causal, mechanistic, externally validated, and reproducible scientific contribution
Primary unit of inference	Canonical physics family; model–family pair for model-specific analyses
Central construct	Model-specific counterfactual shortcut sensitivity under solver-certified answer-preserving cue interventions
Core mechanistic boundary	“Shortcut reliance” is reserved for cases with both behavioural counterfactual evidence and controlled causal intervention evidence
Use of this document	Research execution, experiment review, claim control, go/no-go decisions, paper planning, and audit
Update rule	Changes that alter hypotheses, primary metrics, confirmatory panels, causal sites, thresholds, or claim scope must be entered in the decision log with date, rationale, and consequences

Revision record

Version	Date	Change
1.0	June 22, 2026	Initial Part-II scientific and experimental plan
1.1	June 22, 2026	Corrected benchmark inventory to 155; reclassified donor-on-renamed controls as completed evidence; clarified the residualized correctness result; imposed immediate Wave 0/1 governance; added the operational validator recruitment, assignment, compensation, adjudication, and audit plan

Continuation principle

Part II is not a request to accumulate the maximum number of experiments. Its purpose is to close a falsifiable scientific chain:

verified non-governing cue → specific internal processing

→ localized causal representation → diagnostic hidden-state signature

→ counterfactual instability → selective correction.

Experiments are prioritized according to how directly they close, falsify, or bound one of these links.

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1 Executive Continuation Statement

1.1 Current scientific position

PhysMon has moved beyond proposal-stage plausibility. The project now contains a solver-structured benchmark, measurable model-specific sensitivity, family-held-out prompt-side hidden-state monitors, broad artefact and behavioural baselines, and localized causal intervention results in at least two dense model families. The current evidence supports continued investment. It does not yet support a universal shortcut representation, a complete reasoning circuit, or a claim that all cue families and architectures implement sensitivity in the same way.

The scientifically strongest current interpretation is:

Current bounded conclusion

Prompt-side hidden states contain information about which solver-certified physics families will later exhibit counterfactual output instability. In selected dense models, localized mid-layer attention-head representations can strongly modulate that instability under controlled interventions. The diagnostic representation, causal representation, cue family, and architectural implementation may only partially overlap.

The next phase must therefore answer five deeper questions:

1. **Construct validity:** Are all reported families truly answer-invariant, physically well-posed, naturally worded, and independently validated?
2. **Incremental monitor value:** Do hidden states add predictive information beyond the strongest combination of surface, correctness, confidence, entropy, and generated-text evidence?
3. **Causal structure:** Are the localized components necessary, sufficient, dose-responsive, and specific rather than merely correlated or generically disruptive?
4. **Mechanistic content:** What information reaches the localized components, what computation do they perform, and how does this relate to the diagnostic monitor?
5. **Generalization and control:** Do the findings survive new problem sources, cue compositions, decoding policies, languages, model variants, and selective mitigation?

2 Current Evidence Baseline at the Start of Part II

2.1 Purpose and authority of this snapshot

This section records the empirical starting point that motivates the Part-II programme. Values are approximate summaries of the currently completed artifacts and must be regenerated from the canonical evidence table before public use. Where a corrected or completed artifact supersedes an earlier report, the corrected artifact is authoritative. Partial, zero-row, placeholder, or unfinished runs are not treated as null findings.

2.2 Benchmark and behavioural evidence

The current benchmark contains 155 canonical families after incorporation of the 15-family expansion set. The primary Qwen behavioural sweep summarized below covered the original 140-family benchmark; expansion-family results must be reported separately or recomputed in a fully harmonized 155-family table before public use. On the 140-family Qwen sweep, the current authoritative summaries indicate approximately:

- mean maximum correct-answer log-probability drop near 1.40 nats;
- 70 of 140 evaluated families above the frozen 0.5-nat sensitivity threshold;
- answer parsing coverage near 0.97;

- overall answer correctness near 0.61.

The number 155 describes the benchmark inventory, not a universal denominator for every experiment. Each reported analysis must state the exact eligible family count after model-specific parsing, correctness, sensitivity, and intervention-panel criteria.

The initial answer-flip target was too sparse at pilot scale. The project therefore adopted $S_{\theta}^{\text{lp}}(\tau)$ as the primary continuous target and froze the 0.5-nat binary threshold before the main probe analysis. Part II must preserve this historical sequence and distinguish exploratory metric development from confirmation.

2.3 Monitoring evidence

The strongest current Qwen prompt-side monitoring results are approximately:

Monitor or baseline	Approx. AUROC	Interpretive status
Qwen mean-state hidden-state monitor	0.760	Strongest current primary candidate
Qwen variance monitor	0.731	Historically important discovery path
Ensemble monitor	0.748	Does not clearly dominate mean state
MLP-state monitor	0.742	Useful representation comparison
Held-out-domain macro result	0.749	Borderline pass on domain generalization
No-Cue-C variance result	0.715	Suggests signal is not entirely driven by Cue C
CoT text classifier	0.599	Weak generated-text baseline
Answer-plus-rationale classifier	0.646	Stronger textual baseline
Black-box two-query sensitivity	0.668	Important practical behavioural comparator
Entropy-style proxy	0.727	Strong non-activation competitor
Generic correctness probe	0.766	Strong comparator; raw AUROC slightly exceeds the sensitivity monitor
Sensitivity after correctness residualization	0.738	Positive evidence that the signal survives removal of correctness-related variation
Embedding/earliest-layer representation	0.574	Evidence against a purely lexical first-layer signal

The current monitoring conclusion is useful but not final. Hidden states outperform surface and generated-text monitors, while the margin over uncertainty-related evidence is smaller. The generic correctness probe reaches approximately 0.766 AUROC, compared with approximately 0.760 for the raw sensitivity monitor. However, the sensitivity monitor remains near 0.738 after correctness residualization, showing that the signal does not collapse to generic correctness or difficulty. The remaining construct-validity question is therefore not the raw 0.766-versus-0.760 comparison alone. It is whether the residualized and combined hidden-state signal provides statistically reliable, calibrated, and practically

meaningful information beyond correctness, entropy, confidence, and other non-activation evidence. The approximately 0.022 reduction from the raw to residualized sensitivity monitor must be treated as shared variance rather than ignored.

2.4 Causal and mechanistic evidence

The completed Qwen sweep currently identifies layer 16 head 11 as the strongest localized component on the tested panel. Its normalized recovery is approximately 1.85, with strong effects across the selected target families and substantial separation from random directions and unrelated donors. Head 19 is a weaker secondary contributor. Earlier emphasis on head 26 is superseded by the completed sweep.

The completed Llama analysis identifies layer 21 head 2 as the strongest single-head site, with normalized recovery also approximately 1.85 on the tested panel and successful effects on 14 of 20 target families. Multi-head combinations can produce larger effects, suggesting either partial redundancy, complementarity, or an incompletely localized mechanism.

These results support localized sufficiency-style causal contributions. They do not yet prove a complete circuit or establish that the same representation is necessary, diagnostic, and mechanistically shared across architectures. The values above one further require absolute-effect and denominator analyses.

2.5 Cross-model, cross-cue, and robustness evidence

Current summaries indicate:

- DeepSeek-R1-Distill-Qwen-32B within-model monitoring near 0.703 AUROC;
- Qwen-to-DeepSeek transfer near 0.725 and DeepSeek-to-Qwen transfer near 0.633;
- inconclusive DeepSeek causal replication because the valid model-positive causal subset is too small;
- weak-to-moderate Cue A/B cross-transfer with very low direct direction alignment;
- Cue C showing weaker probing and smaller effects at the Qwen causal heads;
- family-level monitoring near 0.82 on a small expansion set and variant-level performance near 0.77;
- corrected variable-renaming monitor transfer near 0.67, with behavioural and Qwen causal effects remaining present;
- both donor-on-renamed causal controls completed with mean normalized recovery of 0.0; within the tested renamed-control panel this is 100% specificity against variable-name-based donor transfer;
- no support for the proposed H11-to-H15 direct gating relation in the tested design.

The most defensible interpretation is mechanism heterogeneity: similar behavioural vulnerability may leave partially transferable diagnostic signatures while relying on different component-level implementations.

2.6 Incomplete and non-authoritative evidence

The following must not be presented as completed findings until rerun or finalized:

- the incomplete Qwen2.5-3B behavioural sweep;
- absent or unfinished Qwen2.5-14B and Mistral scaling results;
- absent DeepSeek full-head-sweep output;
- stale partial summaries superseded by completed sweeps;
- validation performed by an agent in place of the proposal's required independent human reviewers.

Part-II interpretation rule

The current results are sufficient to justify deeper work. They are not a license to treat every exploratory result as a confirmed contribution. Part II begins by separating authoritative evidence from promising but incomplete signals, and then subjects the central claims to independent validation and untouched confirmation.

2.7 Immediate governance rule: Wave 0 and Wave 1 take precedence

Wave 0 begins immediately. Wave 1 recruitment and validation operations begin in parallel as soon as the benchmark manifest is frozen for annotation. These are scientific prerequisites, not administrative clean-up.

Confirmatory GPU-queue rule

Until the Wave 0 artifact-authority audit and the Wave 1 validation pilot have passed, no newly completed experiment may be designated confirmatory evidence. Exploratory GPU work may continue in isolated directories, but every such run must be labelled `exploratory_unfrozen` and may not be used to change a headline layer, head, threshold, donor policy, benchmark definition, or claimed replication.

The confirmatory queue should pause for new head sweeps, final causal panels, final cross-model replication claims, and final external-transfer experiments. Work that may continue includes infrastructure testing, parser debugging, runtime optimization, path-tracing prototypes, synthetic-task development, and analysis-code development. Foundational work that proceeds immediately includes the experiment registry, canonical model-family table, artifact authority audit, validator recruitment, and the blinded validation pilot.

Required run status	Meaning
<code>complete_authoritative</code>	Completed, internally consistent, and eligible for the frozen evidence base
<code>corrected_authoritative</code>	Corrected result that explicitly supersedes an earlier artifact
<code>complete_exploratory</code>	Completed but used for discovery, selection, or hypothesis generation only
<code>superseded</code>	Replaced by a corrected or completed artifact and prohibited from headline use
<code>partial_not_reportable</code>	Incomplete run whose available rows are not a scientific null or final estimate
<code>failed</code>	Execution or validity failure with documented reason
<code>pending</code>	Planned but not completed
<code>diagnostic_only</code>	Engineering, parser, or instrumentation result not used as substantive evidence

The previously reported overall sensitivity-specific fraction, including any 68.1% summary, must be regenerated from authoritative raw outputs after the renamed controls are incorporated. The document records the local renamed-panel result—0.0 mean recovery in both controls and 100% specificity within that tested panel—but does not infer a revised pooled percentage without an explicit numerator, denominator, and eligibility rule.

2.8 Primary next-phase thesis

The recommended thesis is no longer merely that activations detect shortcut-sensitive model behaviour. The stronger target is:

Recommended central thesis

Solver-certified counterfactual shortcut sensitivity leaves a pre-generation diagnostic signature, but the most diagnostic representation is not necessarily identical to the representation that causally produces the instability. PhysMon will identify the relationship between diagnostic and causal subspaces, trace a partial upstream pathway into the strongest causal components, characterize the cue properties those components process, and test whether the resulting account generalizes across cue classes, model families, and naturally sourced physics problems.

A secondary high-upside thesis is:

High-upside extension

Counterfactual consistency training reduces shortcut sensitivity by suppressing, relocating, or replacing the internal mechanisms that mediate cue dependence. Tracking this change across checkpoints can reveal whether robustness learning removes a mechanism, reroutes it, or merely hides its behavioural expression.

2.9 Non-goals and prohibited claim inflation

The next phase must not claim any of the following without substantially stronger evidence:

- recovery of the model’s complete hidden reasoning path;
- discovery of a universal “shortcut neuron” or universal circuit;
- a shared mechanism across all cue types and all model families;
- causal mediation solely from linear probe performance;
- complete circuit recovery from one or two causally active heads;
- semantic interpretation of raw residual coordinates without basis-invariant evidence;
- successful model editing merely because an intervention changes one benchmark metric;
- general scientific-reasoning robustness from a controlled text-only benchmark alone.

3 Governing Scientific Framework

3.1 Three levels that must remain separate

Every experiment, table, and claim must explicitly identify which level it studies.

Level	Object	Permitted inference
Level 1	Prompt condition, cue value, cue position, rendering, lexical form	A property of the stimulus. Decoding this level does not establish model reliance.
Level 2	Behavioural counterfactual sensitivity measured across solver-certified invariant family members	A model-specific vulnerability or invariance violation. This is the primary monitor target.
Level 3	Internal representation or mechanism whose intervention changes Level-2 behaviour	A bounded causal contribution when supported by necessity, sufficiency, specificity, and damage controls.

3.2 Formal objects

Let a canonical template be

$$\tau = (r, z, a, g, y^*),$$

where r is the set of governing variables, z is the designated cue or non-governing variable, a is the set of assumptions, g is the governing physical law, and y^* is the solver-certified answer.

The invariant family is

$$\mathcal{F}_\tau = \{x_\tau(r, z_i, a)\}_{i=1}^m,$$

with the certified invariance condition

$$y_\tau^*(r, z_i, a) = y_\tau^*(r, z_j, a) \quad \forall i \neq j.$$

The current primary continuous behavioural target is the maximum correct-answer log-probability drop:

$$S_\theta^{\text{dp}}(\tau) = \max_{j \neq \text{base}} [\log p_\theta(y^* \mid x_\tau(r, z_{\text{base}}, a)) - \log p_\theta(y^* \mid x_\tau(r, z_j, a))].$$

Additional targets include answer-flip sensitivity $\widehat{S}_\theta(\tau)$, distributional divergence $S_\theta^{\text{JSD}}(\tau)$, stochastic failure probability, and intermediate-quantity instability.

An internal monitor is

$$f_\phi(h_\theta(x)) \rightarrow \widehat{S},$$

trained on development-family activations and evaluated on held-out canonical families. A causal intervention manipulates an activation, component, direction, or subspace and tests its effect on Level-2 sensitivity.

3.3 Claim hierarchy

Claim tier	Minimum evidence	Allowed wording
Descriptive	Reliable behavioural sensitivity on valid invariant families	“The model is counterfactually sensitive” or “violates expected invariance.”
Predictive	Held-out monitoring beyond appropriate baselines and artefact controls	“Prompt-side hidden states predict later sensitivity.”
Localized causal	Component intervention exceeds random and damage controls	“The component contributes causally to sensitivity.”
Mechanistic function	Competing hypotheses are falsified using controlled factorial interventions	“The component processes or transports property X under the tested conditions.”
Partial circuit	Upstream and downstream paths are verified with knockout, rescue, and path-specific tests	“A partial causal pathway contributes to sensitivity.”
Mitigation	Selective intervention reduces sensitivity while preserving capability on held-out data	“The monitor or mechanism supports selective correction.”

4 Evidence Stabilization and Research Governance

4.1 Master experiment registry

Before additional experiments become part of the paper evidence, create a canonical registry with one row per run. Required fields are listed in Appendix A. Every run receives one of the following statuses:

- `complete_authoritative`;
- `corrected_authoritative`;
- `complete_exploratory`;
- `superseded`;
- `partial_not_reportable`;
- `failed`;
- `pending`;
- `diagnostic_only`.

No figure or paper statement may draw from a run whose status is ambiguous. A correction must never silently overwrite the original record; the registry must preserve both the historical run and the reason it was superseded.

4.2 Canonical model–family evidence table

Construct one authoritative table indexed by (model, canonical family). It should contain all behavioural, probing, baseline, and causal fields needed for final analysis. Rendered prompt variants and repeated generations remain nested under their family rather than being treated as independent observations.

Required columns include:

- family, domain, cue, template, rendering, and split identifiers;
- solver answer, certificate hash, validation status, and parser status;
- correctness and log-probability for every family member;
- all sensitivity metrics;
- primary monitor prediction and calibration score;
- all major baseline predictions;
- eligibility for causal panels;
- raw and normalized intervention effects;
- exclusion flags and reasons;
- version hashes for data, code, model, and configuration.

4.3 Authority rules for conflicting artifacts

Use the following hierarchy:

1. corrected raw outputs with valid provenance;
2. completed raw outputs;
3. completed summaries generated from those outputs;
4. exploratory analysis documents;
5. partial summaries;
6. prose status reports.

A prose report cannot override a later corrected numerical artifact. An empty file, zero-row output, one-byte marker, or incomplete array is never interpreted as a null scientific result.

4.4 Discovery, internal confirmation, and external confirmation

4.4.1 Discovery partition

May be used for choosing metrics, thresholds, layers, heads, probe types, donor rules, and analysis ideas.

4.4.2 Internal confirmatory partition

Must remain untouched during selection. The primary site, direction, threshold, and statistical test must be frozen before evaluation.

4.4.3 External confirmatory partition

Naturally sourced or independently constructed families that are not used for model or analysis selection. These should be evaluated with frozen monitors, causal sites, and thresholds.

Contamination rule

Any family changed after observing model behaviour, probe score, or intervention effect returns to the discovery set. A repaired family may re-enter confirmation only after independent revalidation and only if the repair was not optimized to restore a desired model result.

4.5 Experiment cards and pre-analysis specifications

Every confirmatory experiment should have a one-page card completed before launch:

- scientific question;
- hypothesis and plausible null;
- independent unit;
- inclusion and exclusion criteria;
- model and checkpoint;
- dataset and split hash;
- intervention or monitor definition;
- primary and secondary metrics;
- planned controls;
- exact statistical test;
- success, redesign, and stop criteria;
- permitted claim if successful;
- permitted interpretation if null.

5 Publication-Grade Human Validation

5.1 Objective and operational principle

Human validation must establish that the benchmark's central physical invariance claim is trustworthy independently of model behaviour. It is not sufficient for a symbolic script to return identical values if the natural-language assumptions are incomplete, the cue is actually relevant, or the rendering changes the intended problem. Agent-assisted review may support clerical checking or issue discovery, but it is not a substitute for the independent human protocol defined here.

The validation programme requires named roles, recruited people, blinded assignment, compensation, adjudication, and a frozen audit trail. Validation begins as an operational workstream during Wave 0 rather than being deferred until after further mechanistic experiments.

5.2 Required personnel and ownership

5.2.1 Validation coordinator

Appoint one project member as validation coordinator. The coordinator is responsible for recruitment, screening, validator training, blinded packet creation, assignment balancing, missing-field checks, agreement computation, adjudication routing, compensation records, and final manifest production. The coordinator must not reveal model outputs or expected family status. Where the coordinator helped construct a family, that person must not serve as its deciding adjudicator.

5.2.2 Primary validators

Recruit at least four primary validators. Every confirmatory family receives exactly two independent primary reviews. Four validators are preferred to one fixed pair because this distributes workload, reduces fatigue, allows pairwise agreement checks, and prevents one validator pair from being confounded with a particular cue family or domain.

5.2.3 Adjudicators

Recruit one or preferably two adjudicators with stronger domain expertise. A preferred structure is one adjudicator for mechanics and one for electrostatics/basic circuits. A faculty member, doctoral researcher, teaching assistant, or comparably qualified advanced researcher may serve in this role. Adjudicators see disputed families and anonymized primary justifications, but not model sensitivity, probe scores, or causal results.

5.3 Recruitment mechanism

Recruitment should use direct, documented channels rather than an anonymous crowd pool. Suitable channels include:

- advanced physics students at BITS campuses;
- postgraduate physics, electrical-engineering, or related students;
- teaching assistants for mechanics, electromagnetism, or circuits courses;
- research scholars identified through faculty or laboratory networks;
- recent graduates with demonstrable advanced coursework in the assigned domain.

The initial operational target is to contact 8–10 candidates, screen at least 5–6, appoint four primary validators, and confirm one or two adjudicators. The coordinator maintains a recruitment log containing contact date, qualification evidence, screening outcome, assigned role, conflict declaration, and availability.

5.4 Qualification and screening

A validator should satisfy at least one of the following:

- an undergraduate or postgraduate degree in physics, electrical engineering, mechanical engineering, or a closely related field;
- advanced undergraduate coursework covering the assigned mechanics or electromagnetism/circuits material;
- research or teaching experience demonstrating competence in the assigned domain;
- successful performance on the project qualification test.

Degree title alone is not sufficient. Every candidate completes a 10–15-family qualification test containing valid invariances, subtly relevant distractors, missing assumptions, unit-compatible governing variables, invalid frame transformations, semantic-drift paraphrases, parser ambiguities, and incorrect

symbolic certificates. The test requires short written derivations. Candidates qualify only if they correctly identify the scientifically critical cases, especially cue relevance and answer invariance.

5.5 Independence and conflict declaration

Before assignment, each validator confirms that they:

- did not construct the assigned benchmark families;
- have not seen model sensitivity labels, probe predictions, or intervention results;
- will not coordinate decisions with the second validator;
- will disclose uncertainty rather than infer the project author’s expected verdict;
- will report any prior involvement or conflict of interest.

The paper and benchmark card should state that validation was independent of model outputs.

5.6 Compensation and workload planning

Compensation must reflect the expertise and derivation effort required. It must not depend on agreement with the benchmark authors or on accepting a family as valid.

Use a timed pilot to set compensation:

1. assign 10 representative families to each candidate;
2. record active completion time, including derivation and written justification;
3. estimate median time per family and expected fatigue-adjusted throughput;
4. set hourly or per-family compensation under institutional rules;
5. compensate pilot work where administratively possible, including candidates not selected after screening.

The coordinator records expected and actual workload and avoids assignment blocks large enough to induce systematic fatigue.

5.7 Annotation platform and immutable schema

Use a structured form backed by an append-only or versioned table. The first validator must not be able to view the second validator’s response. Each packet contains an anonymized family ID, all family variants, stated assumptions, solver-derived answer, and solver certificate. It contains no model names, behavioural sensitivity, probe scores, expected validity flag, or causal result.

The stored schema should include at least:

```
validator_id, family_id, domain, assignment_pair,
assigned_at, completed_at, governing_law,
relevant_variables, designated_cue, cue_relevant,
cue_relevance_reason, assumptions_sufficient,
answer_unique, answer_invariant, semantic_equivalence,
wording_natural, difficulty_shift, solver_correct,
parser_answer_correct, overall_verdict, confidence,
free_text_notes, adjudication_status
```

All edits after submission must create a new version rather than silently overwriting the original response.

5.8 Validator training and calibration pilot

Before full annotation, create a 20–25-family blinded pilot stratified across domain, cue type, original versus expansion families, expected difficulty, paraphrase status, variable renaming, and known questionable cases. The pilot includes:

- unambiguously valid irrelevant variables;
- subtly relevant variables incorrectly labelled irrelevant;
- insufficient assumptions;
- ambiguous frames or sign conventions;
- dimensional inconsistencies;
- natural and unnatural renderings;
- paraphrases that change semantics;
- correct and incorrect solver derivations;
- unit-compatible distractors that are actually governing;
- valid and invalid representation transformations.

Validators explain the governing law and cue irrelevance rather than selecting only binary labels. After the pilot, compute agreement, inspect disagreement causes, revise instructions, and repeat calibration where necessary. Full validation of all 155 families begins only after the pilot protocol is stable.

5.9 Balanced assignment design

Use overlapping validator pairs rather than one fixed pair. With validators A, B, C, and D, distribute blocks across A–B, A–C, A–D, B–C, B–D, and C–D. Each pair receives a mixture of domains, cue types, original and expansion families, and difficulty levels. No validator pair should be responsible for only one cue family.

Assignment must be generated before model-result review and recorded in the manifest. Validators work independently, and responses remain hidden from one another until adjudication.

5.10 Family-level validation form

Each validator independently records:

1. governing physical principle and derivation outline;
2. variables entering the governing expression;
3. designated cue and explanation of non-governance;
4. sufficiency of assumptions;
5. uniqueness of the answer;
6. correctness of the symbolic derivation;
7. invariance across every cue value or rendering;
8. dimensional and unit correctness;
9. absence of unintended semantic changes or co-variation;
10. naturalness and approximate difficulty matching;
11. correctness of the canonical answer;

12. parser-normalized answer validity;
13. final verdict: valid, minor repair, or invalid;
14. confidence and written justification.

Critical fields—cue relevance, answer invariance, assumptions, and semantic equivalence—must not be left blank.

5.11 Adjudication and revalidation

A family is sent to adjudication whenever primary validators disagree about cue relevance, answer invariance, sufficient assumptions, solver correctness, semantic equivalence, or overall validity. The adjudicator receives the family, solver certificate, and anonymized explanations from both validators, but no model outputs.

The adjudicator records the final physics derivation, decision, whether repair is possible, the exact repair required, and whether related families require re-audit. A repaired family undergoes fresh independent validation; it does not inherit its earlier approval. Families repaired after model-result inspection return to the discovery partition unless the repair was fully specified without optimizing toward a desired model outcome.

5.12 Agreement analysis

Report at least:

- raw agreement;
- Cohen’s κ for overall validity;
- agreement and κ for cue relevance;
- agreement and κ for answer invariance;
- agreement for assumption sufficiency and solver correctness;
- agreement by cue family and domain;
- confidence-stratified agreement;
- adjudication and repair rates;
- exclusion rates and reasons.

A single pooled κ is insufficient because agreement on easy wording items can conceal disagreement on the core construct. The primary agreement criteria concern cue relevance and answer invariance.

5.13 Validation deliverables

The validation workstream produces:

- recruitment and qualification log;
- validator handbook and qualification test;
- compensation and workload record;
- immutable blinded annotation table;
- agreement report;
- adjudication log;
- repaired-family and exclusion manifests;
- final signed-off benchmark manifest;
- anonymized examples for the paper appendix and benchmark card.

5.14 Operational start and completion gate

The validation coordinator, candidate list, qualification test, and 20–25-family pilot packet should be created during Wave 0. Recruitment begins immediately rather than waiting for later mechanistic experiments.

Human-validation proceed gate

Proceed with final confirmatory claims only if: (i) all confirmatory families have dual independent review; (ii) no unresolved construct-invalid cases remain; (iii) agreement on cue relevance and answer invariance is high and transparently reported; (iv) every repaired family has passed revalidation; (v) qualifications, assignment, compensation, and adjudication are documented; and (vi) the final benchmark manifest is frozen. If agreement is weak for a cue family, that cue family is redesigned, demoted, or removed from the primary paper.

6 Solver, Parser, and Benchmark Integrity

6.1 Machine-readable solver certificates

Every family should include:

- symbolic variables and units;
- assumptions;
- governing equation;
- symbolic solution;
- cue variable;
- proof or executable check of cue invariance;
- dimensional analysis;
- numerical test cases;
- expected answer and tolerance;
- certificate and code hashes.

Where differentiable, certify

$$\frac{\partial y^*}{\partial z} = 0$$

under the stated assumptions. For discrete transformations, use executable equivalence checks.

6.2 Mutation testing

The verifier must reject deliberately invalid mutations:

- making the cue governing;
- removing a necessary assumption;
- changing a sign or frame;
- replacing a unit;
- altering a relevant variable;
- changing the governing law;
- introducing an ambiguity in the question target.

Report mutation-detection rate by error type. A verifier that accepts most invalid mutations is not a meaningful certificate.

6.3 Answer parser audit

Build a manually labelled parser test set spanning:

- decimals, integers, fractions, scientific notation, and symbolic answers;
- equivalent units and algebraic forms;
- multiple numbers in a response;
- rounded values;
- wrong units;
- refusals, hedges, and no-answer outputs;
- answer values embedded in prose;
- model-specific formatting.

Report exact extraction accuracy, normalized-answer accuracy, false acceptance, false rejection, and model-specific error rates. Sensitivity metrics should exclude or explicitly model parser-uncertain outputs.

6.4 Leakage and family split audit

No test family may share with training:

- symbolic template;
- rendering skeleton;
- numerical sibling;
- paraphrase;
- variable-renamed version;
- solver expression;
- cue phrase template.

Use normalized symbolic hashes and template-structure hashes in addition to textual similarity.

6.5 Benchmark balance and confound audit

Audit distributions of:

- domain, cue, answer range, sign, and unit;
- integer versus non-integer answer;
- prompt length, number count, variable count, equation count, and unit-token count;
- base correctness and confidence;
- sensitivity prevalence;
- source or rendering style.

Confounds should be matched, conditioned on, or explicitly reported. Artificial exact balance is not required; unexplained correlation with the target is unacceptable.

7 Behavioural Construct Consolidation

7.1 Metric hierarchy

Use a pre-specified hierarchy:

1. **Primary continuous:** maximum correct-answer log-probability drop $S_{\theta}^{\text{lp}}(\tau)$.
2. **Primary binary:** thresholded $S_{\theta}^{\text{lp}}(\tau)$ using the frozen threshold, with threshold sensitivity analysis.
3. **Secondary discrete:** answer-flip rate $\hat{S}_{\theta}(\tau)$.
4. **Secondary distributional:** Jensen–Shannon or Wasserstein divergence over normalized answer distributions.
5. **Operational:** stochastic probability of counterfactual failure under sampling.
6. **Mechanistic:** instability of solver-certified intermediate quantities.

The initial answer-flip pilot failure and the subsequent adoption of $S_{\theta}^{\text{lp}}(\tau)$ should be documented as metric development, including the exact point at which the threshold was frozen relative to probe training.

7.2 Sensitivity reliability

For every model–family pair:

- compute uncertainty around $S_{\theta}^{\text{lp}}(\tau)$ when log-probabilities are numerically approximated;
- use repeated generations for stochastic metrics;
- quantify label instability around the threshold;
- report continuous results rather than relying only on binary labels;
- distinguish model-ineligible families from stable families;
- document missingness due to parsing or generation format.

7.3 Behavioural strata

Define and analyze:

Stratum	Definition and scientific role
Silent-correct	Final answers remain correct, but correct-answer probability changes materially. Central to pre-failure monitoring.
Silent-CoT	Written rationale appears physically acceptable, but family sensitivity is high. Central to faithfulness claims.
Visible-failure	Answer flips or becomes incorrect. Easier behavioural detection; important but less novel.
Stable-correct	Correct and invariant. Negative control and mitigation safety set.
Stable-incorrect	Consistently wrong but insensitive. Separates generic correctness from shortcut sensitivity.
Ambiguous / uncertain	Parser or sampling uncertainty prevents reliable classification. Exclude or analyze separately.

7.4 Decoding-policy robustness

Evaluate greedy, answer-only, chain-of-thought, constrained numerical output, and sampled decoding at several temperatures and top- p settings. The prompt-side monitor is deterministic; test whether it predicts stochastic failure probability across policies.

Use sequential sampling or a pre-specified sample count sufficient to produce useful family-level confidence intervals. Do not interpret ten noisy generations as a precise probability estimate.

8 Definitive Hidden-State Monitoring Study

8.1 Freeze the primary monitor

Declare one primary monitor with fixed:

- model and layer;
- prompt position;
- residual, attention-output, MLP-output, mean-state, or variance representation;
- feature scaling;
- probe class;
- regularization grid;
- calibration method;
- family split;
- target definition.

Alternative monitors remain comparisons. The primary method must not change across tables according to whichever variant performs best.

8.2 Nested family-level evaluation

Use nested family-level cross-validation:

- inner folds for regularization and feature choices;
- outer folds for held-out evaluation;
- development families for calibration;
- untouched confirmation families for final reporting;
- no sibling variants across folds.

8.3 Reviewer-proof baseline stack

8.3.1 Surface and lexical

TF-IDF, word and character n -grams, cue-span-only classifiers, prompt statistics, sentence embeddings, and earliest-layer representations.

8.3.2 Behavioural and uncertainty

Correct-answer log-probability, entropy, self-consistency, output variance, generic correctness prediction, and black-box counterfactual two-query checks.

8.3.3 Generated text

CoT classifier, answer-plus-rationale classifier, open-weight judge monitor, and process-verifier or critic score.

8.3.4 Combined non-activation baseline

Train a combined model using all non-activation features. This is the correct practical comparator for the incremental value of internal states.

8.4 Incremental information and calibration

Compare:

1. strongest combined non-activation baseline;
2. hidden-state monitor;
3. combined non-activation plus hidden-state monitor.

Report AUROC, AUPRC, log loss, Brier score, expected calibration error, fixed-FPR sensitivity, fixed-recall precision, and risk–coverage curves. Use paired family-bootstrap intervals for all performance differences.

The critical question is not merely which standalone model has the largest AUROC. It is whether hidden states improve prediction and calibration beyond all information available without activation access.

8.5 Correctness, entropy, and difficulty deconfounding

Perform:

- residual sensitivity prediction after correctness and entropy models;
- matched analysis by correctness and confidence;
- prediction among base-correct families only;
- prediction within silent-correct families;
- conditioning on domain, cue, prompt statistics, and answer range;
- multi-task prediction of correctness and sensitivity;
- projection-out or residualization of correctness directions.

8.6 Cue-condition diagnostics

Train auxiliary probes for cue presence, cue value, cue position, cue type, domain, answer, and correctness. Compare direction geometry and test sensitivity prediction after removing cue-condition directions.

A strong result is that cue identity becomes decodable early, while behavioural sensitivity emerges later and remains predictable after cue-condition information is removed.

8.7 Prompt-position emergence

Annotate semantic checkpoints:

1. end of setup;
2. before cue;
3. after cue;
4. after governing information;
5. after assumptions;
6. after question;
7. final prompt token.

Run both position-specific probes and frozen final-direction transport. Use full prompts rather than naively truncating where possible, because causal transformer hidden states already represent each prefix.

Plot cue-condition AUROC, sensitivity AUROC, correctness AUROC, and causal-component activity over semantic position.

9 Core Causal Confirmation

9.1 Primary component hypotheses

Treat the current strongest sites as hypotheses to be confirmed, not immutable truths:

- Qwen: layer 16, head 11;
- Llama: layer 21, head 2;
- cue- and model-specific alternatives discovered only on development panels.

9.2 Necessity tests

For each component and held-out sensitive panel:

- zero ablation;
- mean replacement;
- stable-family replacement;
- direction removal;
- low-rank causal-subspace removal;
- attention-pattern ablation;
- value/output ablation.

Measure $S_{\theta}^{\text{lp}}(\tau)$, answer flips, correct-answer log-probability, stable-family behaviour, ordinary physics accuracy, language quality, and unrelated-task damage.

9.3 Sufficiency tests

Perform:

- sensitive donor into stable target;
- stable donor into sensitive target;
- within-family high-sensitivity into low-sensitivity variants;
- matched cross-family donors;
- cross-domain donors;
- same-answer/no-cue donors;
- unrelated-family donors.

9.4 Graded intervention and reciprocal effects

Use

$$h(\alpha) = h_t + \alpha(h_d - h_t)$$

for negative, fractional, unit, and super-unit values of α . Test monotonicity, sign reversal, saturation, and overshoot. Run both sensitive-to-stable and stable-to-sensitive directions.

9.5 Intervention reporting

Every normalized recovery metric must be accompanied by:

- raw log-probability shift;
- raw sensitivity change;
- median and trimmed mean;

- denominator distribution;
- sign-consistency rate;
- family-level confidence interval;
- analysis excluding small denominators.

9.6 Mandatory controls

Include norm-matched random directions, random heads, random sites, nearby layers, unrelated donors, same-answer donors, stable donors, answer-bias checks, and general-performance damage checks.

9.7 Llama miss analysis

For each non-responsive family:

1. verify Llama-specific sensitivity;
2. verify parser and denominator stability;
3. verify donor eligibility;
4. classify cue/domain subtype;
5. run exploratory alternate-site discovery;
6. construct new matched families for independent confirmation;
7. test redundancy and interaction if a second component emerges.

A site discovered on the misses cannot be reported as confirmatory on those same families.

10 Diagnostic versus Causal Representations

10.1 Objects to estimate

Estimate:

$$\begin{aligned}
 w_{\text{probe}} & \text{ primary diagnostic direction,} \\
 v_{\text{contrast}} &= \mathbb{E}[h \mid \text{sensitive}] - \mathbb{E}[h \mid \text{stable}], \\
 V_{\text{causal}} & \text{ low-rank subspace optimized for held-out intervention effect.}
 \end{aligned}$$

The causal subspace must be learned only on development families under norm, rank, and retain-set constraints.

10.2 Geometry and overlap

Report cosine similarity, principal angles, canonical correlation, overlap by cue and domain, and cross-model alignment. Compare diagnostic overlap with causal overlap.

10.3 Cross-ablation matrix

Intervention	Diagnostic question	Causal question
Remove probe direction	Does AUROC collapse?	Does sensitivity change?
Remove causal direction	Does diagnostic information remain?	Does sensitivity decrease?
Preserve only probe direction	Is it sufficient for prediction?	Is it sufficient for behaviour?

Preserve only causal subspace	Is it diagnostic?	Does it retain causal effect?
Random matched subspace	Baseline for geometry	Baseline for intervention

10.4 Low-rank decomposition

Within H11/H2, identify rank- k subspaces for $k \in \{1, 2, 4, 8, \dots\}$ and test how much of the full-head effect is retained. Patch the subspace and its orthogonal complement separately.

A compact causal subspace would strengthen interpretability and enable controlled mitigation. A full-rank result would indicate distributed computation and should be reported as such.

10.5 Sparse probes

L1 probes may quantify compressibility but are basis-dependent. Use random rotations, fold stability, seed stability, and subspace-level causal tests before drawing mechanistic conclusions.

11 Intervention-Based Mediation

11.1 Target causal chain

Test the proposed chain

$$Z \rightarrow H_{\text{causal}} \rightarrow R_{\text{diagnostic}} \rightarrow Y,$$

where Z is cue manipulation, H_{causal} is the localized component or subspace, $R_{\text{diagnostic}}$ is the monitored representation, and Y is output sensitivity.

11.2 Required interventions

For paired family members, estimate:

- total effect of cue change on output;
- cue effect on causal component;
- causal-component effect on monitor score;
- monitor-subspace effect on output;
- cue effect with causal component fixed;
- cue effect with diagnostic subspace fixed.

Use intervention-based total, direct, and mediated effects with family-level bootstrap intervals. Interpret results as evidence for a tested decomposition, not proof of a unique causal graph.

12 Upstream Circuit and Pathway Tracing

12.1 Scientific objective

The current causal localization is downstream-heavy. The next goal is to identify which earlier components and positions create or transport the information used by H11/H2.

12.2 Receiver definition

Before tracing, freeze:

- receiver head and layer;
- receiver query position;
- receiver object: pattern, value-weighted result, projected output, or residual contribution;
- donor/target construction;
- behavioural endpoint;
- held-out verification panel.

12.3 Candidate discovery

Search earlier layers over:

- attention-head outputs;
- MLP outputs;
- residual-stream nodes;
- cue-token, question-token, and final-prompt positions;
- other semantically aligned template positions.

Use attribution-based methods for scalable candidate discovery, but verify candidates with exact interventions because attribution methods can miss nonlinear or cancellation-dependent paths.

12.4 Receiver-conditioned path patching

For each candidate u :

1. patch u between paired prompts;
2. isolate the route from u to H11/H2 where technically defensible;
3. measure receiver activation change;
4. measure final sensitivity change;
5. compare with the direct $u \rightarrow Y$ effect;
6. compare with layer-, norm-, position-, and component-matched controls.

The target distinction is between

$$u \rightarrow H11 \rightarrow Y$$

and an unrelated parallel route

$$u \rightarrow Y.$$

12.5 Knockout, rescue, and interaction

A convincing upstream node should satisfy:

- knockout reduces H11/H2 natural effect;
- restoration rescues the effect;
- joint ablation shows non-additive interaction when components lie on one path;
- effect generalizes to unseen families.

12.6 Path completeness and claim scope

Report the fraction of the receiver’s causal effect explained, residual effect after path ablation, parallel pathways, cue-specific differences, and domain differences. Use “partial causal pathway” unless the identified graph explains most of the relevant effect and reasonable alternatives have been excluded.

13 Functional Characterization of the Causal Components

13.1 Competing functional hypotheses

H11/H2 may encode or transport:

- numerical salience;
- unit compatibility;
- recent-token salience;

- variable-name similarity;
- answer correlation;
- syntactic attachment to the requested quantity;
- semantic relevance;
- distractor status;
- general uncertainty or difficulty;
- a more abstract cue-dependence feature.

The experiment must attempt to falsify these alternatives rather than naming the component from its benchmark association.

13.2 Controlled cue-factorial design

Manipulate orthogonally:

Factor	Levels
Physical relevance	governing, algebraically redundant, non-governing
Unit compatibility	matching, non-matching, dimensionless
Numerical magnitude	zero, small, typical, extreme
Numerical closeness	far from, near, or equal to governing quantities
Position	early, middle, near question, post-question note
Syntactic prominence	main clause, parenthetical, subordinate note
Lexical emphasis	neutral, emphasized, explicitly “important”
Repetition	absent, once, repeated
Variable naming	meaningful, arbitrary symbol, misleading name
Answer correlation	none, weak, strong, reversed
Value availability	name plus value, name only, value only

Use length-matched filler and matched templates so that deleting or moving a cue does not create a trivial sequence-length signal.

13.3 Head-function decomposition

Separate:

- query–key score changes;
- attention-pattern changes;
- value-vector transport;
- output-projection contribution;
- direct logit attribution;
- cue-token versus question-token pathways.

Measure behaviour, monitor score, component activation, attention pattern, and causal effect for every factorial condition. Mixed-effects models should include canonical family as a random effect.

13.4 Cue degradation and position

Do not treat all modifications as one scalar degradation axis. Report property-specific curves and interactions. The goal is to identify which cue properties are necessary for H11/H2 activation and which properties affect the diagnostic monitor without affecting the causal component.

14 Cross-Cue and Cross-Domain Mechanistic Structure

14.1 Donor–target matrix

Construct a matrix across cue, domain, and control categories. Record raw effect, sensitivity change, sign consistency, monitor-score change, answer bias, and general damage.

14.2 Mechanistic clustering

Cluster the donor–target matrix and test whether mechanisms organize by cue type, physics domain, problem structure, answer type, or model. Use permutation tests for cluster stability.

14.3 Cue-C boundary analysis

Determine whether Cue C represents:

- the same diagnostic signal but another causal site;
- another diagnostic and causal representation;
- an artefact-prone or weak behavioural construct;
- a genuinely separate sensitivity mechanism.

A coherent boundary result is stronger than forcing all cues into a universal mechanism claim.

15 Temporal Reasoning Trajectories

15.1 Teacher-forced aligned trajectories

Construct solver-derived rationales with aligned semantic stages:

1. principle identification;
2. relevant-variable identification;
3. cue rejection or incorporation;
4. substitution;
5. intermediate calculation;
6. final answer.

Extract activations at the same semantic stage across variants. This is the primary temporal study because independently sampled CoTs are not naturally aligned.

15.2 Natural generated trajectories

For free generations, detect and manually validate events:

- first law mention;
- first cue mention;
- explicit cue rejection or use;
- first substitution;
- first intermediate value;
- answer commitment.

Raw token indices such as token 50 must not be treated as equivalent reasoning stages across outputs.

15.3 Intermediate-quantity sensitivity

Use solver-certified intermediate quantities such as net force, acceleration, equivalent resistance, field, potential, momentum, or energy. Test whether cue sensitivity appears in an intermediate computation and is later corrected, or appears only at final answer selection.

15.4 Prompt-level modulation

Compare generic “reason carefully” instructions, governing-law-first scaffolds, relevance lists, explicit cue-irrelevance statements, few-shot distractor-ignoring examples, matched irrelevant demonstrations, and deliberately misleading demonstrations.

Measure whether behaviour changes through suppression of the causal pathway, downstream compensation, or merely altered confidence.

16 External and Cross-Environment Generalization

16.1 Naturally sourced external families

Construct 40–60 independently sourced families from multiple physics benchmarks or educational sources. Preserve natural wording, add or manipulate non-governing cues, certify invariance, and complete dual human validation.

Before evaluation, freeze the monitor, threshold, causal sites, donor rules, and statistical tests. Report high-confidence calibration, source-specific performance, and intervention effects on a pre-specified subset.

16.2 Cross-lingual robustness

Start with Chinese and one European language. Use bilingual physics validation. Compare:

1. direct English-probe transfer;
2. language-specific centering;
3. language-specific probes;
4. aligned cross-language probes;
5. H11/H2 necessity and sufficiency;
6. cross-language donor transfer.

A failed direct probe transfer is not automatically lexical confounding; it may reflect language-specific activation geometry. Behaviour, representation, and mechanism must be interpreted jointly.

16.3 Familiar versus novel-law physics

Create algebraically matched familiar and fictional-law problems with new names and units. Compare sensitivity, monitor transfer, causal-site transfer, donor transfer, and the effect of explicitly stating the law.

This tests whether the mechanism reflects abstract cue vulnerability or familiarity with textbook templates.

16.4 Same-backbone post-training comparisons

Where available, compare base, instruction-tuned, math-tuned, reasoning-distilled, and counterfactually trained variants of the same architecture. This isolates post-training effects from architecture and tokenization.

16.5 Cross-model subspace alignment

Align diagnostic and causal subspaces using low-capacity mappings learned on training families. Test held-out transfer and random baselines. Determine whether diagnostic representations align more readily than causal mechanisms.

17 Multi-Cue Interaction and Adversarial Evaluation

17.1 Multi-cue factorials

Construct no-cue, single-cue, congruent multi-cue, conflicting multi-cue, repeated-cue, and relevant-plus-irrelevant conditions. Estimate interaction:

$$\Delta_{AB} = S_{AB} - S_A - S_B.$$

Test whether single-cue-trained monitors predict unseen combinations and whether new components are recruited.

17.2 Adversarial monitor evasion

Freeze the monitor and search for:

$$\begin{aligned} \text{stealth sensitivity:} & \quad \text{high } S_{\theta}^{\text{lp}}(\tau) \text{ and low monitor score,} \\ \text{false alarm:} & \quad \text{low } S_{\theta}^{\text{lp}}(\tau) \text{ and high monitor score.} \end{aligned}$$

Use controlled mutations, evolutionary search, or model-assisted proposal generation, but retain only solver-certified and human-valid prompts. Analyze whether evasions recruit another causal mechanism.

18 Mitigation and Selective Control

18.1 Monitor-triggered intervention

Use the monitor to decide when to apply:

- stable-direction or stable-activation replacement;
- causal-direction removal;
- explicit relevance analysis;
- symbolic solver escalation;
- additional sampling;
- routing to a stronger reasoning model.

Compare hidden-state triggers with entropy, confidence, correctness probes, random routing, and intervene-on-all policies. Report risk-coverage, compute-accuracy, sensitivity reduction, and unnecessary-intervention rate.

18.2 Counterfactual consistency training

Train from the same base checkpoint under:

1. standard physics SFT;
2. matched-data SFT without invariance supervision;
3. counterfactual consistency training.

A representative objective is

$$\mathcal{L} = \mathcal{L}_{\text{task}} + \lambda \sum_{i < j} D(p_{\theta}(y \mid x_i), p_{\theta}(y \mid x_j)).$$

Use multiple λ values and seeds. Track behavioural sensitivity, monitor AUROC, causal-site strength, diagnostic-causal overlap, site migration, held-out cue transfer, and external-family performance over checkpoints.

18.3 Constrained weight editing

Only after a compact causal subspace is validated, consider a constrained low-rank update to reduce cue sensitivity while preserving retain-set behaviour. A generic ROME-style edit is not assumed appropriate for an attention head. Evaluate collateral damage, mechanism migration, and compensation by other components.

19 Synthetic Ground-Truth Validation

19.1 Controlled shortcut-learning models

Train small decoder-only models on symbolic physics-like tasks where a non-governing variable is correlated with the answer at controlled strengths. Reverse or remove the correlation at test time.

Track behavioural sensitivity, monitor AUROC, causal localization, diagnostic-causal alignment, dose-response, and circuit emergence through training.

19.2 Planted internal route

Insert a known low-rank path from cue representation to output logits. Blind the analysis pipeline to its location. Test whether PhysMon recovers the correct layer, component, direction, intervention effect, and null when the route is disabled.

This calibrates method sensitivity and specificity. It does not prove that the same interpretation is correct in large pretrained models.

20 Optional Sparse Autoencoder Study

Train an SAE only after the component-level causal story is stable, using a broad activation corpus of sensitive physics, stable physics, no-cue physics, numerical non-physics, general language, and unrelated reasoning.

Report reconstruction, sparsity, dead features, dictionary-size sweeps, feature stability, and causal feature validation. A correlated SAE feature is not considered mechanistic until ablation or activation changes sensitivity with appropriate controls.

21 Statistical Analysis Plan

21.1 Independent unit and clustering

The canonical family is the independent unit. Variants, paraphrases, numerical siblings, and repeated generations are nested. Use clustered or hierarchical analyses where appropriate.

21.2 Confidence intervals and tests

Use family bootstrap intervals, paired randomization or permutation tests, hierarchical mixed-effects models, and exact sign tests where appropriate. Report effect sizes, not only p -values.

21.3 Multiple comparisons

Correct exploratory layer, head, token, cue-factor, and direction scans. Confirmation on untouched families should use frozen sites and analyses without reselection.

21.4 Seed stability

Repeat probe training, subspace optimization, sampling, fine-tuning, and SAE training across multiple seeds. Report distributions and failure rates.

21.5 Power planning

Before new confirmatory studies, estimate family counts required for monitor differences, intervention effects, cross-cue transfer, and training-induced changes. Never count rendered variants as independent families.

21.6 Missingness and eligibility

Report missing or ineligible families by reason: parser failure, model format failure, lack of base correctness, lack of model-specific sensitivity, unstable denominator, or invalid benchmark family. Avoid silently changing denominators across models.

22 Ordered Execution Programme and Decision Gates

22.1 Wave 0: evidence stabilization

Tasks:

- master registry;
- canonical model–family table;
- artifact authority audit;
- discovery/confirmation freeze;
- claim–evidence matrix;
- experiment-card system;
- corrected benchmark inventory of 155 canonical families;
- validator coordinator appointment and recruitment launch;
- confirmatory GPU-queue pause and exploratory-run labelling.

Wave 0 gate

Every existing result is traceable to a model, dataset hash, configuration, raw output, summary, and status. No headline claim depends on a partial, stale, zero-row, or superseded artifact.

22.2 Wave 1: benchmark trustworthiness

Tasks: recruit and qualify validators, run the 20–25-family blinded pilot, dual human validation, adjudication, repaired-family revalidation, solver mutation tests, parser audit, leakage audit, balance audit, agreement report, and final benchmark manifest.

Wave 1 gate

All confirmatory families are independently validated and solver-certified. No unresolved construct-invalid cases remain. Cue families failing validity or agreement standards are repaired, demoted, or removed.

22.3 Wave 2: behavioural and monitoring confirmation

Tasks: multi-metric sensitivity, silent-sensitivity strata, definitive nested monitoring, combined baseline, incremental information, deconfounding, calibration, prompt-position emergence, stochastic decoding.

Wave 2 gate

Hidden states provide meaningful held-out information beyond the strongest non-activation combination, especially among base-correct or silent-sensitivity families. If not, reframe the monitor as a compact internal proxy rather than a superior detector.

22.4 Wave 3: core causal confirmation

Tasks: necessity, sufficiency, reciprocal patches, dose-response, absolute metrics, damage controls, Llama miss analysis, model-specific causal panels.

Wave 3 gate

At least one model has strong necessity and sufficiency evidence with controlled damage, and a second model supplies credible replication or a clearly characterized boundary. If necessity fails, retain a sufficiency-only causal claim.

22.5 Wave 4: mechanism and pathway

Tasks: diagnostic-causal subspaces, mediation, low-rank decomposition, upstream tracing, exact path verification, knockout/rescue, cue-factorial function analysis, donor-target matrix.

Wave 4 gate

The project can explain at least one bounded relationship between cue property, upstream processing, causal component, diagnostic signature, and behavioural instability. If no functional account survives falsification, report localization without semantic naming.

22.6 Wave 5: generalization

Tasks: external natural families, multi-cue interactions, decoding robustness, cross-lingual triangulation, novel laws, same-backbone variants, adversarial evasion.

Wave 5 gate

The generalization domain and failure boundaries are empirically known. Claims are scoped to the environments in which frozen monitors and interventions work.

22.7 Wave 6: temporal and mitigation studies

Tasks: aligned reasoning trajectories, intermediate quantities, prompting modulation, monitor-triggered interventions, solver routing.

Wave 6 gate

The monitor or mechanism supports a selective correction policy that improves invariance or reliability more efficiently than confidence, entropy, random routing, or intervene-on-all controls.

22.8 Wave 7: high-upside extensions

Tasks: consistency training, mechanism migration, synthetic planted mechanisms, SAE analysis, constrained editing.

These may become a second paper if they dilute the central static-monitoring and causal-mechanism story.

23 Work-Package Specification

WP	Objective	Required outputs	Primary failure mode	Decision
WP0	Stabilize evidence	Registry, canonical table, authoritative manifest, claim matrix	Conflicting or untraceable outputs	Pause new paper claims until resolved
WP1	Validate benchmark	Dual annotations, adjudication, certificates, parser audit	Cue not truly irrelevant or wording changes semantics	Repair or remove family/cue
WP2	Consolidate behaviour	Multi-metric sensitivity and reliability	Signal exists only under one unstable metric	Narrow construct or redesign variants
WP3	Confirm monitor	Nested held-out evaluation and combined baselines	Hidden states add no incremental information	Reframe as diagnostic compression, not superiority
WP4	Confirm causality	Necessity, sufficiency, dose-response, damage controls	Patch effect is generic, unstable, or donor-biased	Drop strong causal wording
WP5	Separate subspaces	Diagnostic-causal geometry and cross-ablation	Probe direction is non-causal or causal direction non-diagnostic	Make dissociation the result
WP6	Trace pathway	Upstream candidates, path patching, knockout/rescue	No compact upstream path; strong redundancy	Report distributed or partial pathway
WP7	Interpret function	Factorial cue study and head decomposition	Head responds to trivial salience or position	Reinterpret and bound claims
WP8	Generalize	External, cross-cue, stochastic, cross-language studies	Results collapse outside templates	Scope benchmark and monitor narrowly
WP9	Mitigate	Selective intervention and routing	Correction damages stable performance	Do not claim actionable control
WP10	Track learning	Consistency training and mechanism migration	Behaviour improves without interpretable mechanism change	Report robustness effect separately
WP11	Calibrate methods	Synthetic planted-mechanism recovery	Pipeline fails known ground truth	Revise mechanistic methodology

Table 10: Part-II work packages, required outputs, principal failure modes, and decision responses.

24 Risk Register

Risk	Severity	Detection	Mitigation
Human validators reject many families	Critical	Low agreement or cue-relevance disputes	Remove invalid families; redesign cue taxonomy; preserve audit trail
Monitor signal is mostly uncertainty/correctness	Critical	Combined baseline matches hidden states	Focus on silent-correct strata; conditional analyses; reframe contribution
Causal effect is sufficiency-only	High	Necessity interventions fail	State bounded sufficiency claim; avoid “relies on” wording
Head effect is generic confidence amplification	Critical	Similar effects on stable/unrelated prompts	Absolute metrics, damage controls, donor specificity, answer-bias tests
Upstream pathway is distributed	Medium	No compact verified path	Report distributed route; use partial circuit language
Cue C remains distinct or weak	Medium	Poor transfer and separate sites	Treat as boundary or separate mechanism
External transfer is weak	High	Frozen monitor fails natural families	Restrict claim to controlled testbed; study domain adaptation
Cross-lingual transfer fails	Low–medium	English direction does not transfer	Distinguish language geometry from behavioural mechanism
Adversarial evasion is easy	High	Many stealth-sensitive cases	Report monitor limits; adversarial retraining; ensemble monitors
Consistency training moves rather than removes mechanism	Medium	Causal site migrates	Make mechanism migration the result
Multiple-testing inflation	Critical	Site chosen on test data	Discovery/confirmation separation and correction
Incomplete model-specific panels	High	Donors not positive in target model	Build model-specific panels before causal inference

25 Paper Architecture and Claim Freeze

25.1 Recommended main-paper backbone

1. Formal construct and solver-certified benchmark.
2. Behavioural counterfactual sensitivity and silent-vulnerability strata.
3. Pre-generation hidden-state monitoring with incremental information beyond combined baselines.
4. Qwen and Llama causal localization with necessity, sufficiency, dose-response, and controls.
5. Diagnostic versus causal subspace relationship.
6. One deep mechanism section: upstream partial pathway or factorial functional characterization.

7. Frozen external-family confirmation and explicit generalization boundaries.

25.2 Supplementary material

Place complete validator documentation, all baseline variants, full layer/head sweeps, cue-factorial details, stochastic decoding, cross-language results, expanded statistics, and failure cases in the supplement unless essential to the central claim.

25.3 Likely follow-up paper

Counterfactual consistency training, mechanism migration, synthetic ground-truth calibration, SAE analysis, and constrained editing can form a distinct follow-up if they make the main paper incoherent or too broad.

25.4 Claim freeze checklist

Before writing final conclusions:

- freeze authoritative data and result manifests;
- freeze primary analyses and plots;
- document all exclusions;
- list successful, failed, and inconclusive hypotheses;
- map every abstract claim to a table or figure;
- remove causal wording unsupported by necessity/sufficiency evidence;
- ensure each generalization claim names its tested environments;
- distinguish discovery from confirmation throughout.

26 Reproducibility and Release

26.1 Benchmark package

Release canonical templates, rendered variants, solver certificates, validation annotations, adjudications, split manifest, parser, benchmark card, and limitations.

26.2 Code package

Release activation extraction, probe training, baselines, interventions, statistics, plotting, environment lock, exact commands, and cluster templates.

26.3 Figure reproducibility

Every figure and table should regenerate from immutable summary tables and a versioned configuration using one documented command. No manual spreadsheet edits or copied numbers should be required.

26.4 Research transparency

Document model versions, tokenizers, quantization, GPU use, random seeds, human qualifications, annotation compensation, API or local-model use, known benchmark weaknesses, and all corrected or superseded analyses.

27 Definition of Scientific Completion

PhysMon is ready for its strongest paper when the following conditions hold.

27.1 Construct validity

- all reported confirmatory families are solver-certified and independently validated;
- agreement and adjudication are reported;
- parser performance is quantified;

- split and leakage integrity are demonstrated.

27.2 Behavioural validity

- sensitivity exists under multiple aligned measures;
- silent and visible cases are distinguished;
- stochastic and deterministic findings are linked;
- threshold uncertainty is reported.

27.3 Monitoring validity

- evaluation is family-held-out;
- surface and cue confounds are controlled;
- hidden states add information beyond combined non-activation baselines;
- calibration and operating-point metrics are reported.

27.4 Causal validity

- necessity and sufficiency are tested;
- intervention strength has a coherent dose-response;
- random, donor, site, answer-bias, and damage controls are passed;
- causal claims remain model- and cue-scoped.

27.5 Mechanistic depth

- diagnostic and causal representations are compared;
- at least one functional hypothesis about H11/H2 is falsifiably tested;
- an upstream partial pathway is identified or distributed processing is established;
- cross-cue and cross-model boundaries are characterized.

27.6 Generalization and use

- frozen monitors are tested on natural external families;
- decoding-policy robustness is known;
- selected cross-environment tests are complete;
- selective correction is compared with strong routing baselines, if claimed.

Final scientific standard

The project is not complete when every possible experiment has been run. It is complete when the central causal chain has been tested at each link, the strongest alternative explanations have been actively falsified, the remaining limitations are known, and every claim can be reproduced from an immutable evidence base.

A Master Experiment Registry Schema

Field	Required content
Experiment ID	Stable unique identifier
Scientific question	One sentence linked to a Part-II work package

Hypothesis	Directional or falsifiable prediction
Experiment class	Exploratory, internal confirmatory, external confirmatory, diagnostic
Model	Exact repository, checkpoint, revision, tokenizer, dtype, quantization
Code provenance	Git commit, dirty-state flag, environment hash
Data provenance	Benchmark version, split hash, family manifest, exclusions
Target	Behavioural, monitoring, causal, mechanistic, mitigation
Metric	Primary and secondary metrics with definitions
Activation/intervention	Layer, component, position, representation, donor/target policy
Seeds	All random seeds and deterministic flags
Compute	Hardware, batch, job ID, runtime, memory
Status	Authoritative, corrected, exploratory, superseded, partial, failed, pending
Outputs	Raw path, summary path, plots, checksums
Corrections	Links to predecessor/successor runs and rationale
Paper eligibility	Main, supplement, diagnostic only, excluded
Claim supported	Exact bounded claim, or “none”

B Human Validation Form Template

1. Family ID and domain.
2. State the governing physical principle.
3. List all governing variables.
4. Identify the designated cue.
5. Explain whether the cue can affect the requested answer.
6. Are the assumptions complete and unambiguous?
7. Is the answer uniquely determined?
8. Verify the symbolic derivation.
9. Verify dimensional consistency.
10. Verify invariance across every family member.
11. Identify any unintended co-variation.
12. Assess wording naturalness and difficulty matching.
13. Verify canonical and normalized answers.
14. Verdict: valid / minor repair / invalid.
15. Confidence and written justification.

C Confirmatory Experiment Card Template

Question
Hypothesis

Null interpretation
Independent unit
Data and split hash
Model and revision
Frozen site/direction
Primary metric
Secondary metrics
Controls
Inclusion criteria
Exclusion criteria
Statistical test
Success threshold
Redesign threshold
Stop/pivot threshold
Allowed claim if
successful
Allowed conclusion if null

D Claim–Evidence Matrix Template

Claim	Required evidence	Current evidence	Status
Behavioural sensitivity exists	Valid invariant families and reliable output changes		
Hidden states predict sensitivity	Held-out family monitoring and baselines		
Signal is not cue presence	Cue probes, residualization, matched controls		
Signal is not generic difficulty	Correctness/entropy-conditioned analyses		
Component is necessary	Targeted removal with low damage		
Component is sufficient	Donor patching and dose-response		
Mechanism is functionally interpreted	Factorial falsification		
Partial upstream pathway exists	Path patching, knockout, rescue		
Monitor generalizes	Frozen external evaluation		
Monitor supports correction	Selective intervention beats baselines		

E Primary Figure and Table Plan

Item	Purpose	Required content
Figure 1	Construct and pipeline	Three levels, family structure, monitoring and causal questions
Figure 2	Behavioural phenomenon	Sensitivity distributions, silent strata, domain/cue breakdown
Figure 3	Monitoring	Layer/position curves, combined-baseline comparison, calibration
Figure 4	Causality	Qwen/Llama necessity, sufficiency, dose-response, random controls
Figure 5	Diagnostic vs causal	Geometry, cross-ablation, low-rank effect retention
Figure 6	Mechanistic depth	Upstream path or cue-factorial functional result
Figure 7	External confirmation	Frozen monitor and intervention on natural families
Table 1	Benchmark	Families, validation, domains, cues, sources
Table 2	Baselines	AUROC, AUPRC, log loss, calibration, fixed-FPR results
Table 3	Causal effects	Raw and normalized effects with confidence intervals
Table 4	Generalization	Cue/domain/model/language/source transfer

F Statistical Reporting Checklist

For every headline estimate, report:

- exact family count and eligibility rule;
- central estimate and 95% confidence interval;
- mean, median, and sign consistency when relevant;
- unit of resampling;
- random seeds;
- correction for selection or multiple comparison;
- exact baseline and paired difference;
- missingness and exclusions;
- whether analysis was exploratory or confirmatory;
- whether hyperparameters or sites were selected on the same data.

G Methodological References

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